Power BI Guide - Retail Sales Analysis

Honest note: the Power BI file (.pbix) is made in Power BI Desktop, which is not available here. So the file is **not included**. Everything you need to make it in about 1 hour is here: the 4 data files, the model picture, the formulas (DAX) and the page layouts. The two dashboard pictures are layout previews (made with Python), not Power BI screenshots. The DAX formulas were not run in Power BI - test them there.

Step 1 - Load the 4 files

Open Power BI Desktop > **Get data > Text/CSV**. Load these files from the `powerbi` folder: `Fact_Sales.csv`, `Dim_Customer.csv`, `Dim_Product.csv`, `Dim_Date.csv`. Click **Load**.

Step 2 - Make the data model (Day 30)

Open **Model view** and join the tables like the picture. Each join is **one-to-many** (one on the small table, many on Fact_Sales):

One side	Many side
Dim_Customer[Customer_ID]	Fact_Sales[Customer_ID]
Dim_Product[Product_ID]	Fact_Sales[Product_ID]
Dim_Date[Date]	Fact_Sales[Date]

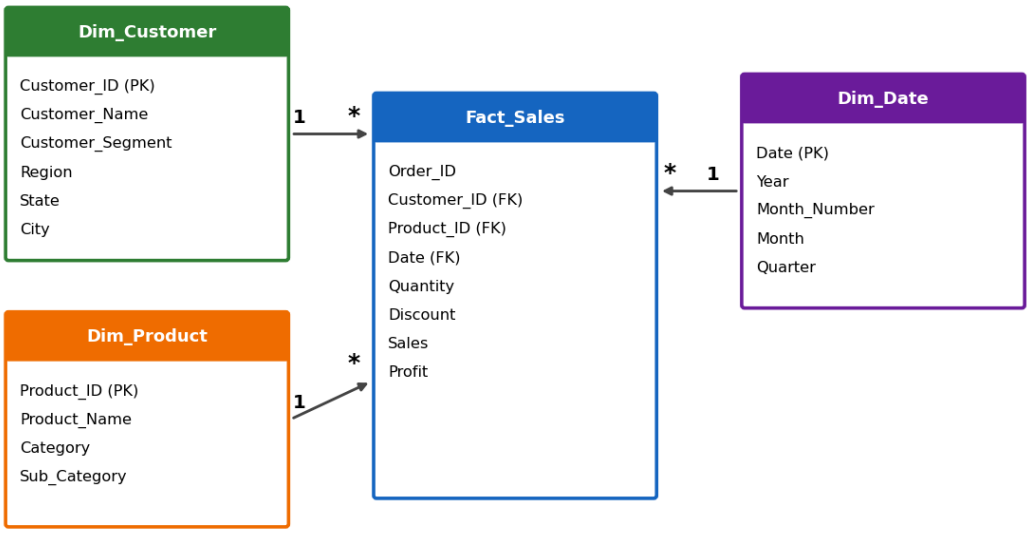


Figure 1: Star schema (Fact table in the middle, 3 dimension tables around it)

- **Fact table** = the numbers (Sales, Profit, Quantity). **Dimension tables** = the details (who, what, when).
- **Primary key** = unique ID in a dimension table. **Foreign key** = the same ID in the Fact table.
- Right-click Dim_Date > **Mark as date table** > choose `Date`. Sort `Month` by `Month_Number`.
- Take a screenshot of Model view and save it as `Data_Model_Screenshot.png`.

Step 3 - Make the measures (DAX)

Right-click Fact_Sales > **New measure** and paste one by one:

```
Total Revenue = SUM ( Fact_Sales[Sales] )
```

```
Total Profit = SUM ( Fact_Sales[Profit] )
```

```
Total Orders = DISTINCTCOUNT ( Fact_Sales[Order_ID] )
```

```
Total Customers = DISTINCTCOUNT ( Fact_Sales[Customer_ID] )
```

```
Average Order Value = DIVIDE ( [Total Revenue], [Total Orders] )
```

```
Profit Margin % = DIVIDE ( [Total Profit], [Total Revenue] )
```

```
Last Year Revenue = CALCULATE ( [Total Revenue], SAMEPERIODLASTYEAR
( Dim_Date[Date] ) )
```

```
Revenue Growth % = DIVIDE ( [Total Revenue] - [Last Year Revenue], [Last Year
Revenue] )
```

```
Avg Spend per Customer = DIVIDE ( [Total Revenue], [Total Customers] )
```

```
Customer Rank = RANKX ( ALLSELECTED ( Dim_Customer[Customer_ID] ), [Total
Revenue] )
```

Check: with no filter your cards must show: revenue ₹12.77 crore, profit ₹1.64 crore, orders 4,396, customers 754 (same as the notebook).

Step 4 - Page 1: Executive Dashboard (Day 31)

- 6 **cards**: Total Revenue, Total Profit, Total Orders, Total Customers, Average Order Value, Profit Margin %.
- **Line chart**: Dim_Date[Month] vs Total Revenue. **Line chart**: Dim_Date[Month] vs Total Profit.
- **Column chart**: Dim_Product[Category] vs Total Revenue. **Bar chart**: Dim_Customer[Region] vs Total Revenue.
- **Top 10 products**: bar chart of Product_Name vs Total Revenue, then Filters > Top N = 10.
- **Slicers**: Year, Month, Region, Category.

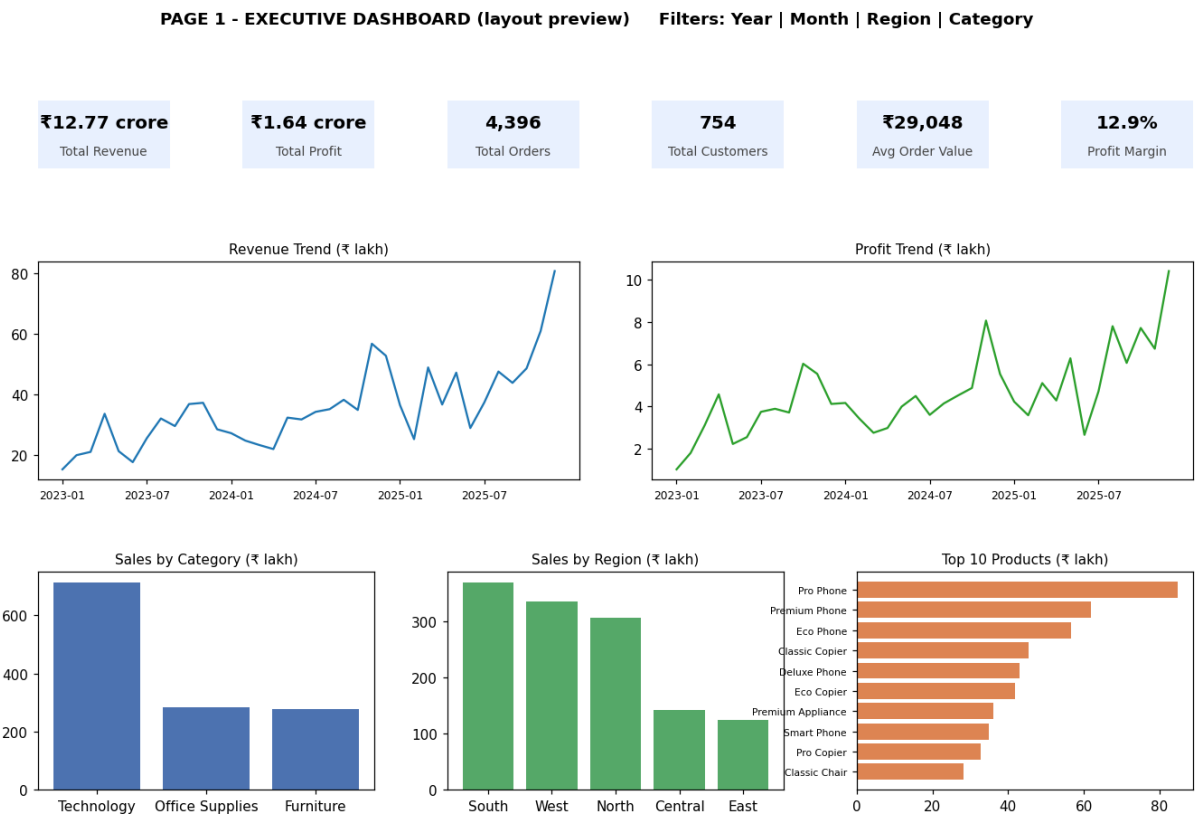


Figure 2: Page 1 layout preview

Step 5 - Page 2: Customer and Product Dashboard (Day 32)

- **Customers:** top 10 customers bar chart, a table with Customer Rank, Total Revenue and Orders, and a chart of Customer_Segment vs Total Revenue.
- **Products:** top 5 and bottom 5 products (Top N filter), Total Profit by product, and category performance (Sales and Profit by Category).
- **Slicers:** Region, Category, Customer_Segment. **Cross-filtering:** click a bar and the other charts change (works by default).
- **Drill-down:** put Category > Sub_Category > Product_Name in one chart axis and use the drill arrows. **Tooltips:** add Total Profit and Profit Margin % in the Tooltips box of the chart.

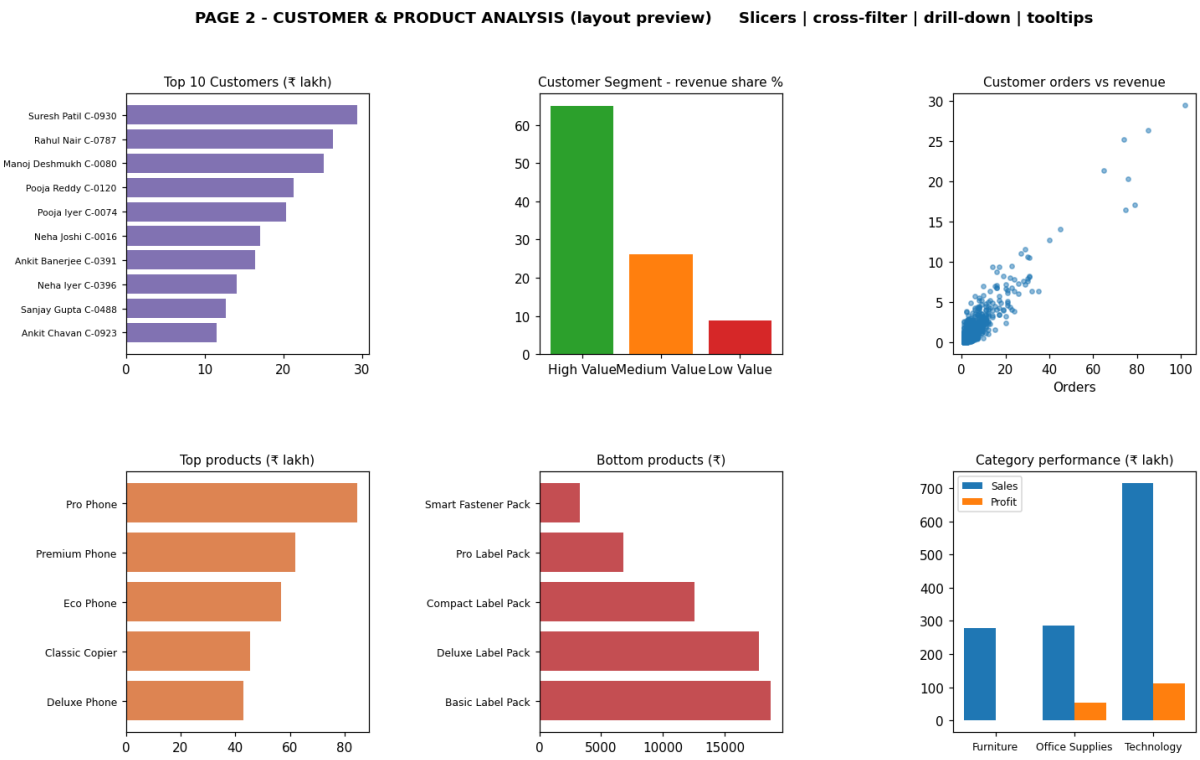


Figure 3: Page 2 layout preview

Step 6 - Save

Save as `Retail_Analytics.pbix`. Export both pages as PDF if your mentor asks for it.